
UNIT 11 TRADITIONAL MODELS OF BUSINESS CYCLES*

Structure

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11.0 OBJECTIVES

After going through this unit you would be able to

- explain the concept and features of business cycle;
- identify the various phases of business cycle;
- ascertain the theoretical framework which explains the occurrence of business cycle;
- appreciate the Keynesian view on business cycle;
- explain the multiplier-accelerator model of business cycle given by Samuelson;
- examine the political business cycle theory;
- appreciate the methodology used by NBER to date the business cycle; and
- distinguish between the leading, lagging and coincident indicators.

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11.1 INTRODUCTION

Economic growth witnessed by a country is smooth; it passes through many ups and downs. There have been periodical ups and downs in the GDP levels of these countries. It is interesting to note that such fluctuation is not limited to economic growth only, it permeates various economic aggregates such as income, employment and prices around their long term trends. Most developed countries have experienced phases of expansion and contraction in output and other economic aggregates. These alternating phases of upswings and downswings are known as business cycles.

You should note that formal analysis of business cycles started in the early 20th century. It was observed that periods of expansion and contraction in an economy occur in a regular manner. The characteristics of these periods are documented by economists such as Wesley Mitchell, Simon Kuznets, and Frederick Mills. Mitchell documented the co-movement of variables over the cycles; Mills documented the co-movement of prices and quantities over expansions and contractions; Kuznets studied the patterns of both growth and fluctuations. The 1930s was a very active period in business cycle research. However, interest in business cycles declined after the publication of Keynes' General Theory which turned attention away from business cycles to short run management of the economy. Interest in business cycles revived in the 1970s when the prevalent economic crisis in many countries could not be explained by Keynesian model.

11.2 FEATURES OF BUSINESS CYCLES

Business cycles consist of recurrent alternating phases of expansions and contractions in a number of economic variables including employment, production, real income, and real sales. Business cycles involve multidimensional processes, in which quantities and prices, stocks and flows, outputs and inputs, real, monetary and financial variables all tend to move together. These are asymmetric in the sense that expansion typically exceeds contractions in size and duration. Business cycles can be distinguished from other fluctuations as they are usually *larger, longer, and widely diffused*.

The major features of business cycles are as follows:

- 1) Though business cycles do not show the same regularity, they have certain distinct phases such as expansion, recession, depression, and recovery. The duration of a business cycle can vary from two years to twelve years.
- 2) Business cycles are *synchronic*. Depression or contraction occurs simultaneously in most industries or sectors of the economy. Recession passes from one industry to another and chain reaction continues till the whole economy is in the grip of recession. Similarly, prosperity spreads through various linkages of input-output relations or demand relations between industries or sectors.

- 3) Fluctuations occur simultaneously in the level of output as well as employment, investment, consumption, rate of interest, price level, etc.
- 4) Consumption of durable goods and investment are affected the most by cyclical fluctuations. As stressed by Keynes, investment is very unstable as it depends on profit expectations of private entrepreneurs. Any change in these expectations makes investment unstable. Thus the amplitude of fluctuation in the case of durable household effects is higher than that of GDP.
- 5) Consumption of non-durable goods and services do not vary much during the different phases of business cycles. Past data of business cycles reveal that households maintain a great stability in the consumption of non-durable goods. Thus the amplitude of fluctuation in the case of non-durable consumption goods is lower than that of GDP.
- 6) The immediate impact of recession or expansion is on the inventories of goods. When recession sets in, inventories start accumulating beyond the desired level. It leads to cut in production of goods. In contrast, when recovery starts, aggregate demand picks up and inventories go below the desired level. It encourages business houses to place more orders for goods which boosts production and stimulates investment.
- 7) Profits fluctuate more than other types of income as business cycles cause uncertainty for the businessmen and make it difficult to forecast economic conditions. During depression, profits turn negative and many businesses go bankrupt.
- 8) Business cycles are international in character. That is, once started in one country, it spreads to other countries through contagion effect. The downslide in financial markets, for example, in one country spreads rapidly to other countries as financial markets are linked globally through capital flows. Further, recession in one country, say the United States (US), can spread to other countries as the imports of the US will decline. Countries which are major exporters to the US will witness a decline in their exports and may witness recession.

11.3 PHASES OF BUSINESS CYCLES

Business cycles are characterized by expansion of economic variables in one period and contraction in the subsequent period. Generally, there are four phases of a business cycle, viz., expansion, recession, depression, and recovery. The four phases of a business cycle are depicted in Fig. 11.1. In the figure the upward sloping straight line indicates the steady state growth path or the long run growth path of GDP. The GDP fluctuates around the steady state growth path due to business cycles. In the expansion phase there is acceleration in growth rate while in the recession phase there is a deceleration in growth rate. In the depression phase economic growth is below its long run trend and the economy can witness negative growth rate also. In fact, negative growth for an economy is not rare. In

2019, several countries (for example, Venezuela, Argentina and Turkey) are having negative economic growth. The difference between recession and depression is one of degree and extent. Similarly, the difference between recovery and expansion is one of degree and extent. After negative growth, the economy passes through the recovery phase and then through the expansion phase. The point at which an expansion ends and a recession begins is called the 'peak' of the business cycle. The point at which a depression ends and recovery begins is called the 'trough'. Thus peak and trough are the turning points in a business cycle.

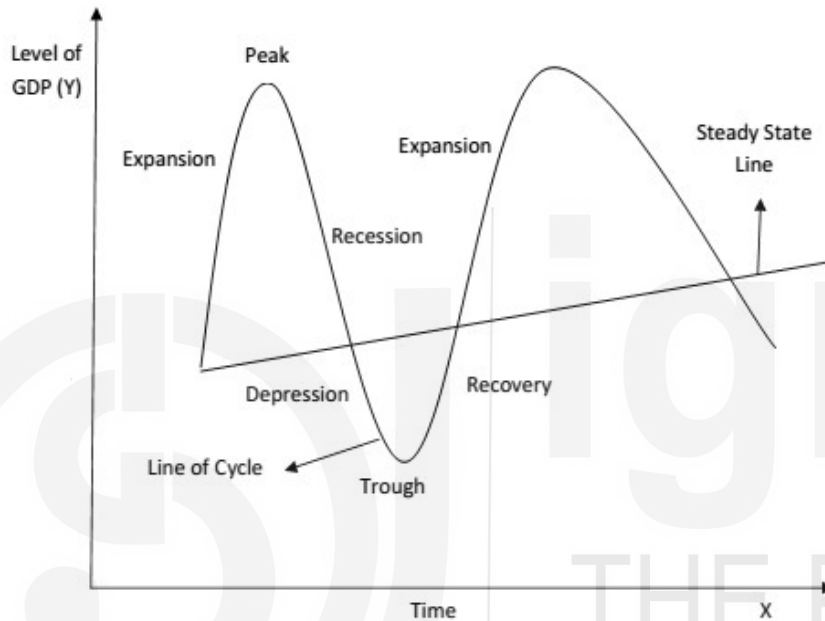


Fig. 11.1: Phases of a Business Cycle

11.3.1 Expansion Phase

In the expansion phase, there is an increase in various economic activities such as production, employment, output, wages, profits, demand and supply of products, and sales. An expansion phase can begin as a result of many forces, including willingness of financial institutions to lend more and willingness of business houses to borrow more. There is overall optimism in the economy. The expansion phase continues till the economic environment is favourable.

During the expansion phase, the economy often gets overheated in the sense that various constraints and frictions develop in the economy. Wage rate and prices increase much faster than output, leading to hike in production cost decline in profits. The central bank also pursues a restrictive monetary policy so that inflation is in control.

Economic growth in the expansion phase eventually slows down and reaches its peak. During the peak of a business cycle, economic variables such as production, profit, sales and employment are high; but do not accelerate further. There is a gradual decrease in the demand for various inputs due to the increase

in input prices. The increase in input prices leads to increase in product prices, while real income of people does not increase proportionately. It leads consumers to restructure their monthly budget and the demand for products, particularly luxuries and consumer durables, starts falling. The peak also occurs before various economic indicators, such as retail sales and the number of employed people, fall. When the decline in the demand for products becomes rapid and steady, 'recession phase' begins.

11.3.2 Recession Phase

In the recession phase, all economic variables such as production, prices, saving and investment start decreasing. Generally, in the beginning of the downturn, producers are not aware of the decrease in demand for their products and they continue to produce goods and services. In such a case, supply exceeds demand and there is accumulation of inventories.

Over time, producers realize that there is an unwanted accumulation of inventories, escalation in production cost, and decline in profits. Such a condition is first experienced by few industries and slowly spreads to the whole economy. During the recession phase, producers usually avoid new investments which lead to the reduction in the demand for factors of production, and consequent decline in input prices and unemployment. Firms reduce levels of production and the number of people on their payrolls. A chain reaction starts, lower income, lower demand, lower output, lower employment, lower income, and so on. The adverse effects of recession extend beyond the purely economic realm and influence the social fabric of society as well. Social unrest and crimes tend to rise during recessions.

During the recession phase, actual output may be below the potential output, thereby indicating a positive output gap. When recession continues further, economic growth rate may be negative also. In such cases, there is not just a decline the growth rate; absolute level of GDP declines. As sales decline, business houses find it difficult to repay their debts. As business sentiments are low enough to carry out new investment, demand for credit declines. Banks also become cautious in their lending, as the chances of default on repayment increases.

The economy however revives its growth rate over a period of time and optimism build up in certain sectors of the economy. This leads to reversal of the recession phase and the 'recovery phase' starts. Individuals and organizations start developing a positive attitude toward the various economic factors, such as investment, employment, and production. In the recovery phase there is an increase in consumer spending and demand for consumer goods. This provides incentives to firms to increase production, carry out new investments, etc. Further, there could be some investment during the recession phase due to replacement of obsolete machineries and maintenance of existing capital stock.

Price mechanism plays a very important role in the recovery phase of economy. As pointed out earlier, during the recession phase decline in input prices is greater than decline in product prices. This leads to a reduction in cost of production and increase in profits. Apart from this, in recovery phase, some of the depreciated capital goods are replaced by producers and some are maintained by them. As a result, investment and employment by organizations increases. As this process gains momentum an economy again enters into the phase of expansion. Thus, a business cycle gets completed.

Check Your Progress 1

1) What is meant by business cycle?

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2) List the important features of a business cycle.

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3) Describe the different phases of business cycle?

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11.4 THEORIES OF BUSINESS CYCLES

An important question is what causes business cycles. Several theories of business cycles have been propounded from time to time. Each of these theories spells out different factors which cause business cycles.

11.4.1 Keynes' Theory of Business Cycle

Keynes in his seminal work 'General Theory of Employment, Interest and Money' made an important contribution to the analysis of the causes of business cycles. According to Keynes, the changes in the level of aggregate effective demand will bring about fluctuations in the level of income. Aggregate demand comprises demand for consumption goods and demand for investment goods. While, propensity to consume being more or less stable in the short run, fluctuations in aggregate demand depend primarily upon fluctuations in investment demand. Multiplier plays a significant role in causing magnified

changes in income following a reduction or increase in investment. However, the theory fails to explain the cumulative character of business cycle. For example, suppose that investment rises by 100 rupees and that the magnitude of multiplier is 4. From the theory of multiplier, we know that national income will rise by 400 and if multiplier is the only force at work that will be the end of the matter, with the economy reaching a new stable equilibrium at a higher level of national income. But in real life, this is not likely to be so, for a rise in income produced by a given rise in investment will have further repercussions in the economy. This reaction is described in the principle of accelerator. Samuelson combined the accelerator principle with the multiplier and showed that the interaction between the two can bring about cyclical fluctuations in economic activity.

11.4.2 Samuelson's Model of Business Cycles: Interaction between Multiplier and Accelerator

Samuelson in his seminal paper convincingly showed that an autonomous increase in the level of investment raises income by a magnified amount depending upon the value of the multiplier. This increase in income further induces the increases in investment through acceleration effect. The increase in income brings about increase in aggregate demand for goods and services. To produce more goods we require more capital goods for which extra investment is undertaken. Thus the relationship between investment and income is one of mutual interaction; investment affects income which in turn affects investment demand and in this process income and employment fluctuate in a cyclical manner.

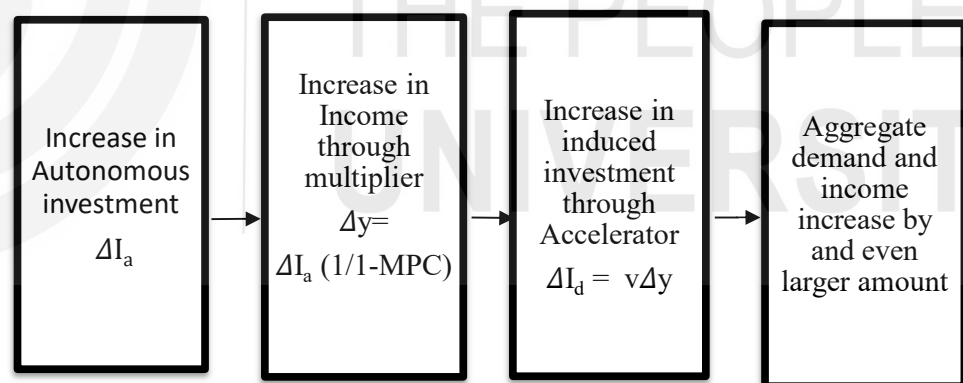


Fig. 10.2 shows how income and output will increase by even larger amount when accelerator is combined with the Keynesian multiplier.

ΔI_a = Increase in Autonomous investment

Δy = Increase in Income

$\frac{1}{1-MPC}$ = Size of Multiplier when MPC = Marginal Propensity to Consume

ΔI_d = Increase in Induced Investment

v = Size of accelerator

The changes in any component of aggregate demand produce a multiplier effect whose magnitude depends upon the marginal propensity to consume. When consumption, income and output increase under the influence of multiplier effect, they induce further changes in investment and the extent to this induced investment in capital goods industries depends on the capital output ratio, that is, the interaction between the multiplier and accelerator without any external shocks can give rise to the business cycles whose patterns differs depending upon the magnitudes of the marginal propensity to consume and capital-output ratio. The model of interaction between multiplier and accelerator can be mathematically represented as under:

$$Y_t = C_t + I_t \quad \dots (11.1)$$

$$C_t = C_a + c (Y_{t-1}) \quad \dots (11.2)$$

$$I_t = I_a + v (Y_{t-1} - Y_{t-2}) \quad \dots(11.3)$$

Where Y_t , C_t , I_t stand for income, consumption and investment respectively for a period t , C_a stands for autonomous consumption, I_a for autonomous investment, c for marginal propensity to consume and v for capital-output ratio or accelerator.

From equation 11.1, we notice that that consumption in a period t is a function of income of the previous period, Y_{t-1} . That is, one period lag has been assumed for income to determine the consumption of a period. Whereas, the induced investment in period t is a function of the change in income in the previous period in equation 11.2. This means that there are two period gaps for changes in income to determine induced investment. In the equation 11.3, induced investment equals $v (Y_{t-1} - Y_{t-2})$. Now, substitute equations 2 and 3 in equation 1 which gives us the equation 11.4.

$$Y_t = C_a + c (Y_{t-1}) + I_a + v (Y_{t-1} - Y_{t-2}) \quad \dots(11.4)$$

This equation reflects how changes in income are dependent on the values of marginal propensity to consume (c) and capital-output ration (v) or accelerator.

In static equilibrium, the level of income can be written as:

$$Y = C_a + cY + I \quad \dots(11.5)$$

This is due to the fact that in static equilibrium, given the data of the determining factors, the equilibrium level of income remains unchanged, that is, in this case, $Y_t = Y_{t-1} = Y_{t-2} = Y_{t-n}$ so that the periods lags have no influence at all and accelerator is reduced to zero as $Y_{t-1} - Y_{t-2} = 0$. Thus, in a dynamic state when autonomous investment changes, the equation 11.4 describes the path which a disequilibrium system follows to reach either a final equilibrium state or moves away from it. But whether the economy moves towards a new equilibrium or

deviates away from it depends on the values of marginal propensity to consume and capital-output ratio.

By taking different combinations of the values of marginal propensity to consume and capital-output ratio, Samuelson has described different paths which the economy will follow. The various combinations of the values of marginal propensity to consume and capital-output ratio are shown in Fig. 11.3.

Fig. 11.3: Combinations of c and v

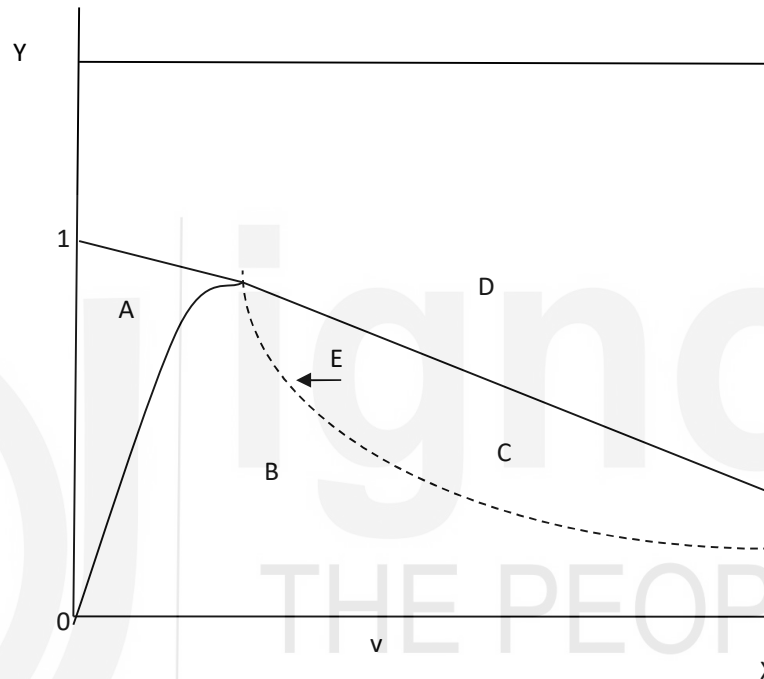
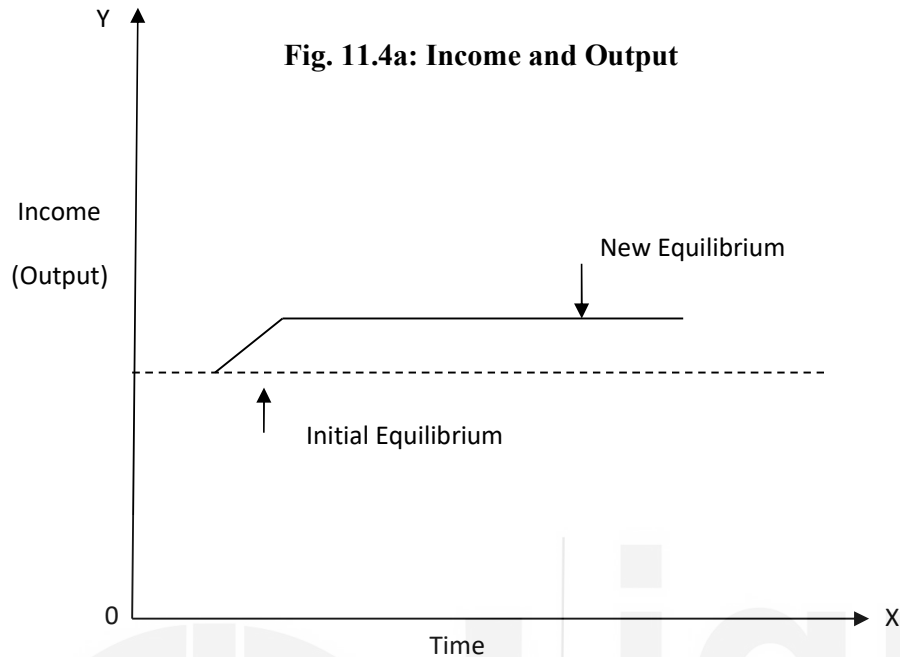


Fig. 10.3 shows the four paths which the economic activity can have depending upon combinations of the values of marginal propensity to consume (c) and capital-output ratio (v).

The four paths or patterns of movements which the economic activity (as measured by gross national product or income) can have depending upon various combinations of the values of marginal propensity to consume and capital-output ratio are depicted in Fig. 11.4 a to 11.4e. When the combinations of the value of marginal propensity to consume and capital-output ratio lie within the region marked as A, with a change in autonomous investment, the gross national product or income moves upward or downward at a decreasing rate and finally reaches a new equilibrium as is shown Fig. 11.4a.



If the values of c and v are such that they lie within the region B, the change in autonomous investment or autonomous consumption will generate fluctuations in income which follow the pattern of a series of damped cycles whose amplitudes go on declining until the cycles disappear as is shown in Fig. 11.4a.

The region C in Fig. 11.3 represents the combinations of c and v which are relatively high as compared to the region B and determine such values of multiplier and accelerator that bring about explosive cycles, that is, the fluctuations of income with successively greater and greater amplitude. The situation is depicted in Fig. 11.4c.

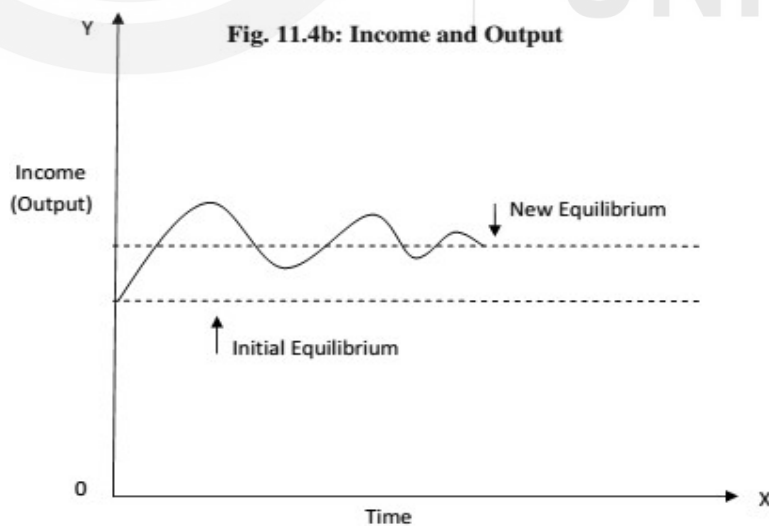
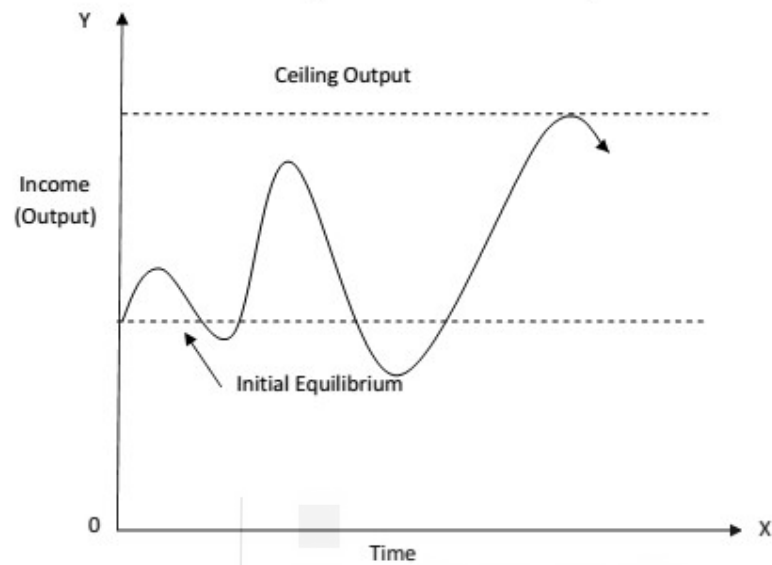
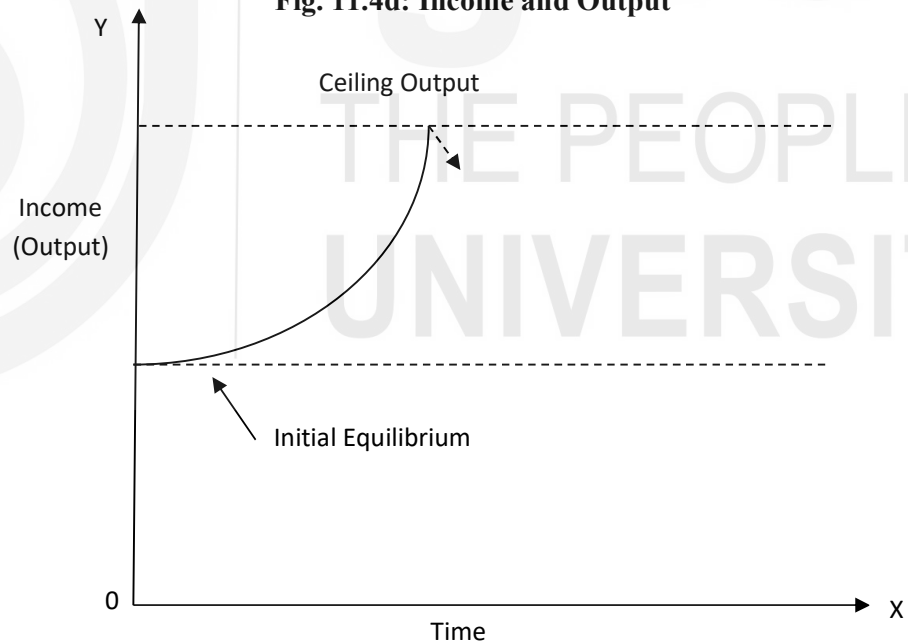


Fig. 11.4c: Income and Output



The region D provides the combinations of c and v which cause income to move upward or downward at an increasing rate. This is depicted in Fig. 11.4d.

Fig. 11.4d: Income and Output



In a special case when values of c and v lie in the region E, they produce fluctuations in income of constant amplitude and are shown in Fig. 11.4e. It is worth noting that all the above five cases do not give rise to cyclical fluctuations or business cycles. It is only combinations of c and v lying in the regions B, C and E that produce business cycles.

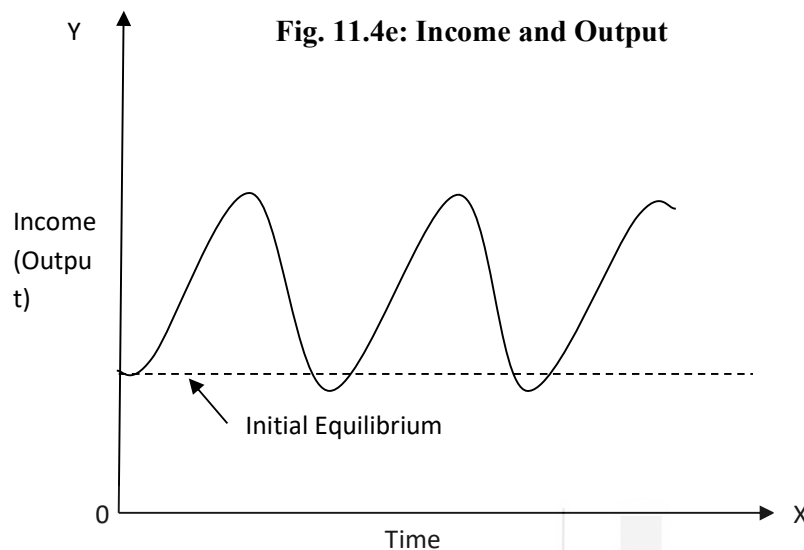


Fig. 11. 4 Panel (a) to (e) shows different patterns of income (output) movements for various values of c and v which respectively determine the magnitudes of multiplier and accelerator.

11.4.3 Political Business Cycles

The term *political business cycle* is used mainly to describe the stimulation of the economy just prior to an election in order to improve prospects of the incumbent government getting re-elected. Expansionary monetary and fiscal policies have politically popular consequences in the short run, such as falling unemployment, economic growth, and benefits from government spending on public services. However, the same policies, especially if pursued to excess, are found to have unpleasant consequences in the long term, such as accelerating inflation and damaging the foreign trade balance. Thus, they can harm the long-term growth potential of the economy. Governments prefer to pursue popular expansionary monetary and fiscal policies for a short horizon immediately before an election. However, being aware of its adverse effects, they do not continue with those measures after they get elected. Thus, after the election is over, government often reverses earlier decisions, which may include cutting spending, slowing the growth of money supply, and allowing interest rates to rise. As a result, the regular holding of elections will produce cyclical fluctuation of economic activity because of recurring patterns of government stimulus and restraint in order to induce an artificial boom in the election time.

11.5 DATING OF BUSINESS CYCLES

Neutralizing the effect of business cycles, particularly the downslides, is a key objective of macroeconomic policy. Understanding recessions and expansions of business cycles is essential for designing of proper frameworks for stabilisation. This requires identifying the chronology of business cycle turning points. In the

US, the National Bureau of Economic Research (NBER) has a dedicated research program for identifying the dates of business cycle turning points. Similarly the Centre for Economic Policy Research (CEPR) Euro Area Business Cycle Dating Committee establishes the chronology of recessions and expansions of Euro Area member countries. In India there have been some attempts to determine the chronology of business cycles.

The NBER's Business Cycle Dating Committee maintains a chronology of the US business cycle. The chronology comprises alternating dates of peaks and troughs in economic activity. A recession is a period between a peak and a trough, and an expansion is a period between a trough and a peak. The NBER does not define a recession in terms of two consecutive quarters of decline in real GDP. Rather, a recession is a significant decline in economic activity spread across the economy, lasting more than a few months, normally visible in real GDP, real income, employment, industrial production, and wholesale-retail sales. Similarly, during an expansion, economic activity rises substantially, spreads across the economy, and usually lasts for several years. Thus, the NBER approach identified cycles as recurrent sequences of alternating phases of expansion and contraction in the levels of a large number of economic time series. This working definition, in substantially its present form, has been in use at the NBER for over fifty years and is currently employed by the NBER to identify and date the US business cycle. These dates are widely accepted by government, academic, and business analysts. The definition served as the basis for the NBER's pre-World War II business cycle chronologies for England, France, and Germany as well as the US.

In both recessions and expansions, brief reversals in economic activity may occur—a recession may include a short period of expansion followed by further decline; an expansion may include a short period of contraction followed by further growth. The Committee applies its judgment based on the above definitions of recessions and expansions and has no fixed rule to determine these upturns and downturns.

The Committee does not have a fixed definition of economic activity. It examines and compares the behaviour of various measures of broad activity: real GDP measured on the product and income sides, economy-wide employment, and real income. The Committee also may consider indicators that do not cover the entire economy, such as real sales and the Federal Reserve's index of industrial production (IIP).

11.6 ECONOMIC INDICATORS

The objectives of macroeconomic policy have long included the avoidance of protracted recessions in which resources go underutilized, and of periods of unsustainable growth that jeopardize reasonable price stability. An important part of the job of the central bank is therefore to gather information on the current state of the economy and if possible, forecast future economic conditions. The

theoretical concept of measuring current business activities using economic series such as GDP, sales, investment, stock prices, is rather simple though its practical application is difficult. Usually, the time pattern of these fluctuating economic series is diverse. While some economic series are expanding at a given point in time, others have already reached their upper turning point and still others are on the downswing; a few economic activities might even be at a lower turning point. Thus the issue arises is how to measure the aggregate state of the economy using these various individual economic series as they have diverse timing patterns.

Economic indicators were conceived at the NBER originally by Mitchell and Burns. This approach requires monitoring of economic variables that tend to be sensitive to cyclical changes no matter what their cause. Burn and Mitchell studied a group of about 487 economic variables to see if turning points in the variables persistently led, coincided with, or lagged turning points in the US business cycle. Seventy one series were chosen and ordered according to the average lead or lag with regard to the reference revivals. For example, six time series had no average lead or lag. On the average, the leading series were from one to ten months ahead of the reference revivals. The lagging series were on the average from one to twelve months behind. The idea of leading, lagging, and coinciding (reference) series was developed even further by G.H. Moore in 1950. Out of 801 series, he chose 21 series which passed a test for conformity and timing; conformity meaning the consistency with which the series have conformed to business cycles and timing referring to the consistency with which a time series' turning point has led, lagged or roughly coincided with the reference data. Those series were selected which were similar in timing at peaks and troughs and were classified into three groups: leading, roughly coincident, and lagging.

11.6.1 Leading Indicators

Leading economic indicators help us in assessing where the economy is heading. They indicate what is coming, such as moving from a peak phase into a contraction, before it actually happens.

One of the most significant leading indicators is the stock market itself, gauged by an index such as the S&P 500. It will begin to rise before economic conditions seems favourable, and it will be begin to decline before economic conditions seem to warrant it. Another important leading indicator is interest rates. Low interests stimulate borrowing and buying, which favours the economy. An increase in interest rates shows the economy is doing well, but eventually rising interest rates lead to a slow down because less people borrow money to start new projects.

11.6.2 Lagging Indicators

Unlike leading indicators, lagging indicators shift after the economy changes. Although they do not typically tell us where the economy is headed, they indicate how the economy changes over time and can help to identify long-term trends. Lagging economic indicators reveal past information about the economy.

The GDP is how much a country is producing. There is significant lag time between when the data is compiled and when it is released, yet it is still an important indicator. Many consider a recession to be underway if two quarters see back-to-back declining GDP. Other indicators, such as the Consumer Price Index (CPI), are also sometimes considered lagging indicators, since they reveal information that is already known to most consumers.

11.6.3 Coincident Indicators

Coincident indicators (more or less) in parallel with the general economic conditions and therefore reflect the current status of the economy. They give consumers, business leaders, and policy makers an idea about where the economy is currently, right now. When the economy rises today, then coincident indicators are also rising right now. And when the economy declines today, then coincident indicators are also declining right now. Typical examples of coincident indicators are industrial production or turnover.

The three types of indicators i.e., leading, lagging and coincident indicators under five categories namely, Investment, Consumption, Employment, Prices and Money are presented in Table 11.1.

Table 11.1: Indicators of Business Cycle		
Leading	Roughly Coincident	Lagging
<i>I. Investment in Fixed Capital and Inventories</i>		
New building permits; housing starts; residential fixed investment New business formation New capital appropriations; contracts and orders for plant and equipment Change in business inventories	Production of business Equipment; Machinery and equipment sales	Backlog of capital Appropriations; Business expenditures for new plant and equipment; and trade inventories

<i>II. Consumption, Trade, Orders, and Deliveries</i>		
New orders for consumer goods and materials; Change in unfilled orders, durable Goods; Vendor performance (speed of deliveries); Index of consumer sentiment	Production of consumer Goods; and trade sales	
<i>III. Employment, Production, and Income</i>		
Average workweek; overtime Hours; Accession rate; layoff rate; New unemployment insurance Claims; Productivity (output per hour); Rate of capacity utilization	Non-agricultural employment; unemployment rate; GNP; personal income; industrial production	Average duration of unemployment; long term unemployment
<i>IV. Prices, Costs, and Profits</i>		
Bond prices; Stock prices; Sensitive materials prices; Profit margins; Total corporate profits; net cash flows		Unit labour costs; Labour share in national income
<i>V. Money, Credit, and Interest</i>		
Monetary growth rates; Change in liquid assets; Change in consumer credit; Total private borrowing; Real money supply	Velocity of money	Short-term interest rates; Bond yields; Consumer credit outstanding; Commercial and industrial loans outstanding

Note: The selection is based on the US indicators published in *Business Conditions Digest*, a monthly report by the Bureau of Economic Analysis, US Department of Commerce

11.7 EMPIRICAL ANALYSIS

Researchers have faced a many of challenges while studying business cycles. One of the major challenges is the problem of representing complex features of economic data. Kuznets, Mitchell, Burns and many others employed various techniques (i.e., moving averages, piecewise trends, etc.) to study the fluctuations of variables around the long term path. As you know, there are various techniques of data analysis. However, there is no single correct way to represent these components.

During the nineteenth century researchers focused on real variables such as agricultural output to explain business cycle. The classical economists assumed perfectly competitive markets and flexibility in prices and wage rate; thus ruled out the possibility of business cycle. According to them, money was neutral in its effects. Later on, during the twentieth century, economists ascribed monetary variables such as money supply, interest rates and prices important variables causing business cycles.

D. K. Backus and P. J. Kehoe in their study titled, “International Evidence on the Historical Properties of Business Cycles” in 1992 took a group of macro variables, in addition to the GDP, for analysis of business cycle. They examined the cyclical behaviour of private consumption expenditure, fixed investment, government purchases and net exports in ten countries, namely, Australia, Canada, Denmark, Germany, Italy, Japan, Norway, Sweden, the United Kingdom, and the US. They find a great deal of regularity in the cyclical behaviour of real quantities. The patterns of variation in all the variables, however, are not similar. The magnitude of output fluctuations varies across countries and over time. Further, the relationships among macroeconomic variables are remarkably stable. For example, investment is about 2 to 4 times more volatile than output while the variation in consumption is almost the same as the variation in output. Both investment and consumption are strongly procyclical. Trade balance is generally countercyclical, exhibiting larger deficits during booms than during recessions. The exception to this regularity in quantities is government purchases, which exhibit no systematic cyclical tendency.

I. Konstantakopoulou and E. G. Tsionas in their work on “Half a century of empirical evidence of Business Cycles in OECD countries” in 2014 examine the existence of common business cycles in OECD countries using H-P filter over the period from 1960 to 2010. The results suggests that there are two distinct cycles in the OECD countries: the Euro area cycle, which includes the business cycles of Germany, France, Italy, Netherlands, Australia and Belgium and the world cycle, which consist of the business cycles of the US, Canada and the U.K.

For a long time, the prevailing view was that we need different theories to explain the different features of the observed data – growth theory to explain the long run trend in a series and something else to explain the cyclical movements

of data. Modern business cycle theory is based on a different premise – the same theory should be capable of explaining both the features. Researchers typically follow one of the two procedures to represent the cyclical component in the data. Some assume a stochastic trend and first difference the (logarithms of the) data to remove it. Most of the authors use a technique known as the Hodrick-Prescott filter (the H-P filter) for representing growth and business cycle components.

Check Your Progress 2

- 1) Give a brief account of Keynesian view on business cycle.

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- 2) Derive the equilibrium income through the model of interaction between multiplier and accelerator.

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- 3) What is meant by conformity and timing of economic series?

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- 4) What is the importance of a lagging indicator?

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11.8 LET US SUM UP

In this unit we focused on three issues, viz., basic features of business cycles , traditional theories of business cycles and measurement of business cycles. Business cycle should be thought of as apparent deviations from a trend. Certain economic variables move together with aggregate output. An examination of the time path of output for any modern industrialized economy reveals that output tends to fluctuate about a long term growth path. These fluctuations about trend

are what we most often think of as the business cycle. The fluctuations are typically irregularly spaced and of varying amplitude and duration. Nevertheless, the one very regular feature of these fluctuations is the way variables move together. Business cycle is characterised by four phases, viz., expansion, recession, depression and recovery. Some of the important mathematical theories which explain these feature of business cycles are: a) Keynesian theory which showed that changes in the level of aggregate effective demand bring about fluctuations in the level of income, output and employment; b) Samuelson model of business cycle which convincingly proved that it is the interaction between the multiplier and accelerator that gives rise to cyclical fluctuations in economic activity, and c) Political theory of business cycle which suggests that business cycles are an outcome of the periodicity of elections in a democracy that prompts the governments to increase public investment before elections and cut down on it afterwards. NBER uses the working definition of business cycle formulated by Burns and Mitchell in the 1940s to identify and date the US business cycle. These dates are widely accepted by government, academic and business analysts. The main problem encountered in the research of business cycle is the lack of a single sufficiently long and consistent measure of aggregate economic activity. The NBER, therefore, relies on the evidence from a large number of comprehensive indicators rather than any single one to date a business cycle. The variables include quantities and prices, stocks and flows, outputs and inputs, real monetary and financial variables. Timing and amplitude of these variables are used to group them into leading, lagging and coincident indicators.

To conclude, a business cycle is not just a movement along a curve in a textbook. It implies new jobs for people, or the loss of them. It is new income, or the loss of it. It is the funds to build new infrastructure such as schools, hospitals and roads. The story of the business cycle is the story of progress and plenty of failure and sacrifice. The effects of recessions extend beyond the purely economic realm and influence the social fabric of society as well. Misery of households, poverty, unemployment, property related crimes, etc. tend to rise during recessions.

11.9 ANSWERS/HINTS TO CHECK YOUR PROGRESS EXERCISES

Check Your Progress 1

- 1) Business cycle consists of recurrent alternating phases of expansion and contraction in a large number of economic activities.
- 2) Refer to Sub-Section 11.2 for the important features.
- 3) Describe the phases given at Section 11.3. You should explain Fig. 11.1 also.

Check Your Progress 2

- 1) Keynes emphasized on fluctuation in aggregate demand. Investment multiplier amplified the magnitude of change in output. Refer to Sub-Section 11.4.1.

- 2) Explain the Fig. 11.2. Derive the equilibrium income through the equations (11.1), (11.2) and (11.3) to reach at the equation (11.4). Refer to Sub-Section 11.4.2.
- 3) Conformity is the consistency with which the series have conformed to business cycles. Timing refers to the consistency with which a time series' turning point has led, lagged or roughly coincided with the reference data.
- 4) Refer to Sub-Section 11.6.2. The importance of a lagging indicator is its ability to confirm that a pattern is occurring. Unemployment is one of the most popular lagging indicators. If the unemployment rate is rising, it indicates that the economy has been doing poorly.



UNIT 6 REAL BUSINESS CYCLES*

Structure

- 12.0 Objectives
- 12.1 Introduction
- 12.2 New Classical View on Business Cycle
- 12.3 Real Factors vs. Monetary Factors
- 12.4 A Baseline RBC Model
- 12.5 Inter-temporal Substitution in Labour Supply
- 12.6 Impact of Supply Shocks to the Economy
- 12.7 New-Keynesian View on Business Cycle
- 12.8 Let Us Sum Up
- 12.9 Answer/Hints to Check Your Progress Exercises

12.0 OBJECTIVES

After going through this unit you would be in a position to

- describe the salient features of new-classical economics;
- outline the assumptions and methodology adopted by new-classical economics on analysis of business cycles;
- bring out the important features of real business cycle theory;
- explain the inter-temporal substitution in labour supply;
- assess the role of supply shocks; and
- discuss the new-Keynesian approach to business cycle.

12.1 INTRODUCTION

In the previous Unit we mentioned that business cycles are synchronic in the sense that the state of recession or boom is prevalent in most industries at the same time. Further, certain variables are pro-cyclical while some others are counter-cyclical. Examples of pro-cyclical variables are output, employment, wages and consumption expenditure. Examples of counter-cyclical variables are unemployment, company profits, and stock market prices.

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There is no consensus among economists about the reasons behind business cycles. According to Keynes (1936) fluctuations in investment (and therefore employment) is caused by ‘animal spirit’, a sort of human emotion. Fluctuation in output gets amplified due to the effect of ‘investment multiplier’. Samuelson (1939) showed that the interaction of multiplier and accelerator leads to business cycles. According to Austrian economist Joseph A. Schumpeter (1961), business cycles are an outcome of the process of innovation and technological change; a process called ‘creative destruction’. Schumpeter emphasises on the role of entrepreneurs, who undertake innovation – they bring in new technology which disrupts existing production processes. Such disruptions in an economy lead to substantial fluctuations in employment and output. There have been several other theories of business cycles which we have not included in the previous Unit. Richard Goodwin (1967), another heterodox economist, ascribes business cycles to the fluctuations in the share of workers in total output. During the boom phase of business cycle, rising employment leads to higher wages, rise in bargaining power of workers, and increase in the share of wages in revenue. This leads to decrease in profit share, decrease in accumulation of capital, and consequent decline in employment. Eventually the rise in unemployment leads to fall in the share of wages in revenue, consequent recovery of profits, and increase in accumulation. In sharp contrast to Keynes, Milton Friedman considered the changes in money supply to be the major reason behind business cycles. According to Friedman, a free market economy is inherently stable. The fluctuation in money supply causes business cycles in an economy.

New classical economists followed the monetarist tradition and emphasized on monetary factors. They introduced rational expectations and microfoundation in macroeconomic analysis. According to them monetary shocks causes business cycles in an economy. The real business cycle theory, however, brings back the importance of real factors such as productivity shocks to explain business cycle.

In this Unit we begin with new-classical view on business cycle and subsequently move on to real business cycle theory. We will conclude the Unit with a note on new-Keynesian view on business cycle.

12.2 NEW-CLASSICAL VIEW ON BUSINESS CYCLE

During the 1950s and 1960s Keynesian economics was very popular – its policy prescription was acceptable to researchers as well as policy makers. Milton Friedman, however, challenged Keynesian economics during that time. According to Friedman, changes in aggregate income are explained mostly by monetary policy. Had the government taken recourse to open market operations and redeemed bonds and securities during the Great Depression, money supply would have increased and the adverse impact would have been lesser. Further, Friedman disagreed with the inflation-unemployment trade-off as suggested by the Phillips Curve. He introduced expectations into the analysis, and pointed out that the Phillips curve is vertical in the short-run. The trade-off is possible only in

the short-run; in the long-run, unemployment cannot be brought down below the natural rate of unemployment.

The onset of stagflation during the 1970s further consolidated the dissent on Keynesian economics. New-classical economics, led by Robert Lucas and Thomas Sargent, brought in a revolution in macroeconomics. New-classical economists revived the classical ideas; the analytical tools however were different. Thus, new-classical economics is completely opposed to the Keynesian policy prescriptions.

New-classical economics is based on three assumptions: (i) rational expectations hypothesis, (ii) continuous market clearing, and (iii) the aggregate supply hypothesis. People have rational expectations; they can correctly anticipate the effect of government actions; and they take counter-measures to neutralise the impact of government policy. A major inference drawn by new-classical economics is that policy measures taken by the government are ineffective. Only unexpected policy shocks lead to temporary deviations in output and employment from their natural levels. The second assumption states that markets clear at any point of time. An implication of the above is the equality between demand and supply, and the economy is in a state of equilibrium always. New-classical economics assumes flexibility in wage rate and prices. Further, adjustments in prices and wages are instantaneous. You may be wondering how the labour market can clear (i.e., there is no unemployment)! This in fact is a debatable assumption of new-classical economics – according to new-classical economists, unemployment is a voluntary phenomenon. At the ongoing wage rate anybody who is willing to work gets employed. The third assumption states that the supply of labour by workers and the supply of output of firms depend upon relative prices. This is the famous Lucas supply function discussed in Unit 7.

Three prominent features of new-classical economics are as follows: (i) microfoundation, (ii) dynamic stochastic general equilibrium, and (iii) quantitative calibration of the economy. Microfoundation means that macroeconomic phenomena are explained through the behaviour of microeconomic agents such as households and firms. For example, the decision of a household is based on its utility function while that of a firm is based on a production function. New-classical models are based on dynamic stochastic general equilibrium (DSGE). As the name suggests, these are ‘general equilibrium models’ in the sense that they pertain to the whole economy. The word ‘dynamic’ reflects the idea that these are inter-temporal models and economic agents optimise on their decision-variables over time. Thus, DSGE models involve the issue of ‘dynamic optimisation’ (See MEC-203, Block 7). As you know, the economy receives various types of shocks, viz., demand shocks, supply shocks and policy shocks, which are exogenous. These shocks are assumed to be random in nature; thus these models are ‘stochastic’. In simple DSGE models there are three inter-related blocks (set of equations), viz., (i) a

demand block, (ii) a supply block, and (iii) a monetary policy equation (for example, policy rule). The equations included in these blocks are derived from micro-foundations.

In order to describe the economy, new-classical economists use the method of calibration. The DSGE models require solution to a set of inter-related dynamic optimisation problems. Here households and firms make decisions on the basis of current and expected future paths for prices and policy variables. This is a challenging task as data are not easily available for longer time periods. Therefore, in DSGE models researchers make extensive use of numerical iteration, so as to replicate the economy. The researcher chooses (from past studies) a set of parameters to describe the utility function and production function. In the process, they rely on model simulation and take parameter values that best fit the empirical data. Such a procedure of comparing the model output with empirical data is called calibration. Remember that calibration is different from estimation used in econometrics.

A question arises at this point: Why do employment and output fluctuate in an economy, if there is market-clearing all the time? In other words, why do business cycles occur? According to new-classical economics, there are two basic reasons: (i) inter-temporal substitution between labour and leisure, and (ii) Lucas supply function. If the actual real wage rate is higher than the expected real wage rate, the household substitutes leisure by labour – thereby working for higher number of hours in a day. In the process it supplies more labour. Thus, there is an increase in supply of labour, which leads to an increase in output (higher than the potential output). As pointed out above, changes in employment are treated as voluntary choices of workers – the amount of labour supplied by households depends upon their rational expectations about real wages. The ‘Lucas Supply Function’ suggests that firms suffer from signal extraction problem and mistake a general price rise as a relative price rise. When there is a general price rise, there is no relative increase in prices of the products produced by a firm, as the cost of production also has gone up. However, firms often see that there is an increase in the prices of their products and they increase their production. Thus, there is a (positive) deviation in employment and output from their natural levels. This leads to a boom or expansion phase in the economy.

Let us take the opposite scenario. Suppose, households perceive that the actual wages they receive is less than expected wage. As a response, households will work less (enjoy more leisure) and supply less labour to the market. Decrease in supply of labour will result in a (negative) deviation in output from its natural level. If the firms perceive that there is a relative decrease in prices of their products, they will decrease their output level. Such decline in output will lead to a recession in the economy. Thus, while the economy is in equilibrium, there could be fluctuation in employment and output.

Check Your Progress 1

- 1) What are the basic assumptions of new-classical economics?

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- 2) Explain the reasons put forth by new-classical economists for occurrence of business cycles in an economy.

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12.3 REAL FACTORS VS. MONETARY FACTORS

Real business cycle (RBC) theory originated with the publication of a paper in 1982 by Kydland and Prescott entitled, “Time to Build and Aggregate Fluctuations”. Kydland and Prescott stated that business cycle is not due to fluctuation in monetary variables; rather it is caused by real shocks to the economy. The RBC theory is a distinct class of new-classical macroeconomics. Therefore, the basic assumptions and analytical tools of new-classical models are applied in RBC models also. In fact, Kydland and Prescott were the first to characterise the general equilibrium of a full-fledged dynamic and stochastic macroeconomic model.

There could be two types of shocks to the economy: (i) nominal shocks, and (ii) real shocks. The nominal shocks (shocks to money supply or money demand) affect the LM curve. The real shocks (shocks to the production function, real government spending, or to savings and consumption decisions) affect the IS curve. In other words, real shocks affect the goods market equilibrium and the labour market equilibrium.

As the name suggests, the RBC models ascribe ‘real shocks’ as the prime reason behind business cycles. The shocks to the production function are known as supply shocks or productivity shocks. Supply shocks can arise due to the following reasons:

- (i) development of new products or production methods (production technology)
- (ii) changes in the quality of labour or capital
- (iii) changes in the availability of raw materials (relative prices, oil crisis)
- (iv) unusually good or bad weather (drought, flood, earthquake)

- (v) changes in government regulations affecting production (tax rate, subsidies, etc.)

The positive or beneficial productivity shocks result in economic booms, whereas the negative or adverse productivity shocks cause recessions in an economy. Let us discuss the effect of a temporary adverse supply shock on the economy (see Fig. 12.1). There will be a decline in the marginal productivity of labour, which will reduce the demand for labour. Consequently, there will be a fall in the real wage. Households will see that their real wage has declined and they will supply lower quantity of labour. This will lead to a fall in the level of output (consequently, a recession). A beneficial productivity shock, on the other hand, would result in an economic boom. There would be temporary hike in productivity in the economy. This will cause an increase in investment and work effort in the current period. Consequently, both the real output and consumption spending will be increased. Once the effect of the supply shock dies down, the economy will be back to natural rate of employment and output.

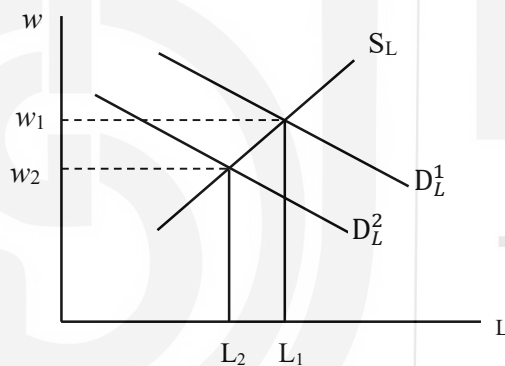


Fig. 12.1: Effect of Temporary Adverse Supply Shock

The RBC theory is based on two basic principles: (i) money has little role to play in business cycles, and (ii) business cycles occur as a response of rational economic agents (firms and households) to real shocks. The economy is continuously at equilibrium – the supply of labour and output are according to rational expectations of economic agents.

The RBC theory has certain differences with new-classical economics. There are at least three differences between the real business cycle theory and the new classical school of thought: (i) the RBC theory does not distinguish between short-run and long-run. It integrates theory of economic growth with business cycle theory. (ii) Lucas advocated that households and firms get fooled – they cannot extract correct signal from price changes in the economy. This results in deviation of actual output from potential output. The RBC theory assumes that households and firms have complete information. (iii) RBC theory assumes that supply shocks arise from real factors while new-classical economics assumes that shocks arise from monetary factors.

The implications of RBC theory are as follows: (i) The government should not try to stabilise output level; it should try to stabilise prices (i.e., inflation should be kept under control – inflation targeting). (ii) The government should not influence aggregate demand through fiscal policy measures (taxes and subsidies), as fiscal measures may not be effective (due to the issue of Ricardian equivalence). (iii) The government should use fiscal policy to stimulate the supply side of the economy (to provide positive supply shocks to the economy).

Real business cycle is criticised on certain grounds: Its assumption of instantaneous market clearing is unrealistic. Prices and wages are rigid in an economy. Further, there is little evidence of large aggregate technology shocks in an economy.

12.4 A BASELINE RBC MODEL

Let us discuss how the fluctuations of aggregated macroeconomic variables can be explained through a general equilibrium model. Recall from microeconomic theory that the Walrasian model (a baseline general equilibrium model) is a competitive model without any of the following: (i) asymmetric information, (ii) externalities, (iii) missing markets, and (iv) other imperfections. If we can explain macroeconomic fluctuations through a Walrasian model, there is no fundamental departure from conventional microeconomic analysis. Real-business-cycle theory aims to analyse how a Walrasian model provides a good explanation of the nature and character of observed fluctuations.

The Ramsey model (see Unit 8) can be seen as a baseline general equilibrium model. The model excludes market imperfections; and there is no heterogeneity among households. It is based on microfoundation in the sense that it considers a representative household and a representative firm. The representative household consumes goods and services which are homogenous. Goods and services produced in the economy are used for consumption by households as well as investment by firms. The representative household supplies labour to the firm. The household determines the quantity of labour to be supplied on the basis of a utility function. Recall that in the Ramsay model an economy converges to a balanced growth path and then grows smoothly.

Let us try to extend the Ramsey model to include aggregate macroeconomic fluctuations. In the absence of shocks, the economy is on a steady state. Therefore, we need to incorporate certain disturbance to the model.

Suppose that there is certain disturbance or shock to the economy. What will be the impact of such shocks? Will the impact of the shock dissipate (die down) after some periods or will it generate fluctuations in output in the economy? The sources of shocks to an economy could be of two types: (i) there could be a shock to the economy's technology. The production technology of the economy undergoes change. Such shocks will lead to a change in the production function across periods. (ii) Change in government purchases also could be a source of disturbance to the economy. These two types of shocks are real disturbances; they are not monetary or nominal disturbances. The change in technology leads

to change in the amount of output produced from a given quantity of inputs. Similarly, the change in government-purchases leads to change in the quantity of goods that are available to the private economy for a given level of production. Since the shocks to an economy are real factors, the models are known as *real-business-cycle* (RBC) models.

We need to make a second modification in the Ramsey model – we allow for change in employment. In the traditional business cycle theories, labour supply is assumed to be exogenous; labour supply is either constant or growing smoothly. As mentioned above, the RBC models are based on microfoundation such that labour supply is determined by a utility function. Households allocate their time between work and leisure – an increase in the time allocated to work implies an increase in the supply of labour and a decrease in the quantity of leisure.

Let us assume that the economy consists of a large number of homogenous price-taking firms and households. There are two inputs to the production function, viz., capital (K) and labour (L). Let us assume that technology is exogenous and it is given by A_t for time period ' t '. Further, let us assume that the functional form of the production function is Cobb-Douglas.

Thus, output in period t is

$$Y_t = A_t F(K_t, L_t) = K_t^\alpha (A_t L_t)^{1-\alpha} \quad \dots (12.1)$$

where $0 < \alpha < 1$

We know that output is divided among consumption (C), investment (I), and government purchases (G). If δ is the depreciation rate of capital in each period, the capital stock in period $(t + 1)$ is

$$\begin{aligned} K_{t+1} &= K_t + I_t - \delta K_t \\ &= K_t + Y_t - C_t - G_t - \delta K_t \end{aligned} \quad \dots (12.2)$$

(Since $C_t + I_t + G_t = Y_t$; we have $I_t = Y_t - C_t - G_t$)

Let us consider that labour and capital are paid their marginal products. Thus, the real wage and the real interest rate in period t are

$$\begin{aligned} w_t &= (1 - \alpha) A_t K_t^\alpha (A_t L_t)^{-\alpha} \\ &= (1 - \alpha) A_t (K_t / A_t L_t)^{-\alpha} \end{aligned} \quad \dots (12.3)$$

and,

$$r_t = \alpha (A_t L_t / K_t)^{1-\alpha} \quad \dots (12.4)$$

(You can derive equation (12.3) and equation (12.4) from equation (12.1) by taking partial derivatives with respect to L and K respectively)

The representative household maximizes the expected value of the utility function

$$U = \sum_{t=0}^{\infty} e^{-\rho t} u(c_t, 1 - l_t) \frac{N_t}{H} \quad \dots (12.5)$$

where, $u(\bullet)$ is the instantaneous utility function. Here, ρ is the discount rate, which represents time preference of utility (i.e., the rate at which future income is discounted). Further, N_t is population and H is the number of households. Then, the average number of members in each household is N_t/H . If the exogenous growth rate of population is n :

$$\ln N_t = \bar{N} + nt, \quad \dots (12.6)$$

where $n < \rho$

Thus, the level of N_t is given by $N_t = e^{\bar{N} + nt}$

Notice that $u(\bullet)$ has two factors in equation (12.5): (i) consumption per member of the household (c_t), and (ii) leisure per member (l_t). Leisure is the difference between the time endowment per member (normalized to 1 for simplicity) and the amount each member works. As all households are homogenous, $c = C/N$, and $l = L/N$.

Let us first discuss about the assumption on the technology shocks. To capture trend growth, we assume the following functional form for technology:

$$\ln A_t = \bar{A} + gt \quad \dots (12.7)$$

where g is the rate of technological progress. Technology in the above specification is not subjected to random shocks. In order to include random disturbances in technology we consider the following functional form:

$$\ln A_t = \bar{A} + gt + \tilde{A}_t \quad \dots (12.8)$$

where $\tilde{A}$ is the random disturbances. The term $\tilde{A}_t$ follows first-order autoregressive process. That is,

$$\tilde{A}_t = \rho_A \tilde{A}_{t-1} + \varepsilon_{A,t}, \quad -1 < \rho_A < 1 \quad \dots (12.9)$$

where, the $\varepsilon_{A,t}$ is a stochastic error term (white noise). It means that $\varepsilon_{A,t}$ has zero mean and zero correlation with other variables in the equation. Hence, equation (12.9) indicates that the random component $\tilde{A}_t$ is a fraction ρ_A of the previous period's value and a white noise error. The effect of a shock to technology disappears gradually over time.

We follow a similar functional form for government purchases: It has a trend component and a white-noise component. There is a trend growth rate of per capita government purchases, but over time government purchases may become arbitrarily large or arbitrarily small relative to the economy. Thus,

$$\ln G_t = \bar{G} + (n + g)t + \tilde{G}_t, \quad -1 < \rho_A < 1, \quad \dots (12.10)$$

$$\tilde{G}_t = \rho_G \tilde{G}_{t-1} + \varepsilon_{G,t}, \quad -1 < \rho_A < 1, \quad \dots (12.11)$$

where the $\varepsilon_{G,t}$ is white noise. This completes the description of the RBC model.

12.5 INTER-TEMPORAL SUBSTITUTION IN LABOUR SUPPLY

We noted in the previous section that major differences between RBC model and the Ramsey model are: (i) the inclusion of leisure in the utility function, and (ii) the introduction of randomness in technology and government purchases as the sources of macroeconomic aggregate fluctuations.

Let us focus on the introduction of leisure in the utility function. For simplicity assume that the household lives only for one period and has only one member, which has no initial wealth. Now, the objective function of the household is $\ln c + b \ln(1 - l)$ and its budget constraint is $c = wl$. Hence, the Lagrangian maximization problem is:

$$\zeta = \ln c + b \ln(1 - l) + \lambda(wl - c) \quad \dots (12.12)$$

The first-order conditions for c and l are as follows:

$$\frac{1}{c} - \lambda = 0, \quad \dots (12.13)$$

$$-\frac{b}{1-l} + \lambda w = 0, \quad \dots (12.14)$$

From equations (12.13), we find that $\lambda = \frac{1}{c} = \frac{1}{wl}$ (since $c = wl$ as per the budget constraint).

Now, substituting the value of λ in equation (12.14), we obtain

$$-\frac{b}{1-l} + \frac{1}{wl} w = 0$$

That is,

$$-\frac{b}{1-l} + \frac{1}{l} = 0 \quad \dots (12.15)$$

The labour supply that satisfies equation (12.15) is independent of the wage rate as wage rate is not included in this equation. The reason is that utility is logarithmic in consumption and the household has no initial wealth. The income effect and substitution effect of a change in the wage rate offset each other. This is the static case, as we assumed that the household lives for one period.

If we consider the case where the household lives for two periods or more than two periods, then the wage variations do affect labour supply. Let us continue to assume that household has no initial wealth with one member who lives for two periods. Also, assume that there is no uncertainty about the interest rate or the second-period wage.

The household's lifetime budget constraint is now

$$c_1 + \frac{1}{1+r} c_2 = w_1 l_1 + \frac{1}{1+r} w_2 l_2 \quad \dots (12.16)$$

where r is the real interest rate. The Lagrangian maximization problem is:

$$\zeta = \ln c_1 + b \ln(1 - l_1) + e^{-\rho} [\ln c_2 + b \ln(1 - l_2)]$$

$$+\lambda[w_1 l_1 + \frac{1}{1+r} w_2 l_2 - c_1 - \frac{1}{1+r} c_2] \quad \dots (12.17)$$

In equation (12.17), the choice variables of the household are c_1, c_2, l_1 and l_2 . Our aim is to show the responsiveness of the relative labour supply due to the variation in relative wage in the two periods. Hence, we need to find out the first-order conditions for l_1 and l_2 from equation (12.17). These conditions are:

$$\frac{b}{1-l_1} = \lambda w_1 \quad \dots (12.18)$$

$$\frac{e^{-\rho} b}{1-l_2} = \frac{1}{1+r} \lambda w_2 \quad \dots (12.19)$$

In order to simplify the above two equations, we divide both sides of (12.18) by w_1 and both sides of (12.19) by $w_2/(1+r)$. Thus, we obtain

$$\frac{b}{1-l_1} \frac{1}{w_1} = \lambda \text{ and } \frac{e^{-\rho} b}{1-l_2} \frac{1+r}{w_2} = \lambda \quad \dots (12.20)$$

$$\text{Now, } \frac{e^{-\rho} b}{1-l_2} \frac{1+r}{w_2} = \frac{b}{1-l_1} \frac{1}{w_1}$$

The relative labour supply in the two periods is:

$$\frac{1-l_1}{1-l_2} = \frac{b}{e^{-\rho} b(1+r)} \frac{w_2}{w_1} = \frac{1}{e^{-\rho}(1+r)} \frac{w_2}{w_1} \quad \dots (12.21)$$

In the two-period case, the wage rate enters into the labour supply function. Equation (12.21) implies that there is a relation between the relative labour supply and the relative wage. This is to note here that $(1-l_1)$ means the household releases l_1 that means the remaining $(1-l_1)$ refers to the leisure in the first period. If w_1 rises relative to w_2 , then l_1 has to be increased or $(1-l_1)$ has to be decreased to satisfy equation (12.21). That means if the wage rate increases in the first period relative to the second period, labour supply has to increase in the first period relative to the second period (or leisure has to decrease in the first period relative to the second period). The elasticity of substitution between leisure in the two time periods is 1 (due to its logarithmic functional form).

Let us consider the case of a rise in r in equation (12.21). It indicates that there is an increase in the labour supply in the first period relative to second-period. A rise in interest rate in the first period raises the incentive to work more (hence, earn more and save more) in the first period relative to the second period. You might have observed that the effect of interest rate is crucial to fluctuation in employment in an economy. The effect of relative wage and interest rate on the labour supply (or employment or leisure) is known as intertemporal substitution in labour supply.

Check Your Progress 2

- 1) Give four examples of negative real shocks to an economy?

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- 2) Explain the concept of intertemporal substitution of labour.

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- 3) Explain how relative labour supply is dependent upon relative wage.

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- 4) In an intertemporal utility function, what is the impact of a rise in interest rate on labour supply?

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12.6 IMPACT OF SUPPLY SHOCKS TO THE ECONOMY

Now let us discuss the impact of the productivity shocks or technology shocks through our RBC model as we discussed in the previous section. Assume that there is an exogenous and temporary (lasting for one time period) favourable shock to technology in a given time period (t). This is reflected in our model by a rise in the A_t term as in equation (12.1). Let us denote the initial level technology as A_{0t} which increased to A_{1t} due to the shock. The growth in technology will cause a rise in the income Y_t at the given levels of K_t and L_t .

The effect of this shock is illustrated in Fig. 12.2. Initially, at the level of technology A_{0t} the corresponding production function is $K_t^\alpha (A_{0t} L_t)^{1-\alpha}$. Let us assume that L_0 is the optimal amount of labour at this level of technology and corresponding equilibrium output is Y_0 . The beneficial productivity shock (or technology shock) shifts the production function upward to $K_t^\alpha (A_{1t} L_t)^{1-\alpha}$. In addition to this upward shift, the nature of the shock is assumed to be such that the production function becomes steeper for any level of labour input. We know from microeconomics that the slope of the production function is the marginal product of labour. Thus, the positive technology shock increases marginal productivity also. Hence, at the same level of labour input (L_0), we have higher output Y'_1 due to the increase in productivity. That means the favourable shock has changed the production possibilities as well. If we increase the labour input to L_1 , the corresponding higher output is Y_1 due to the increased productivity.

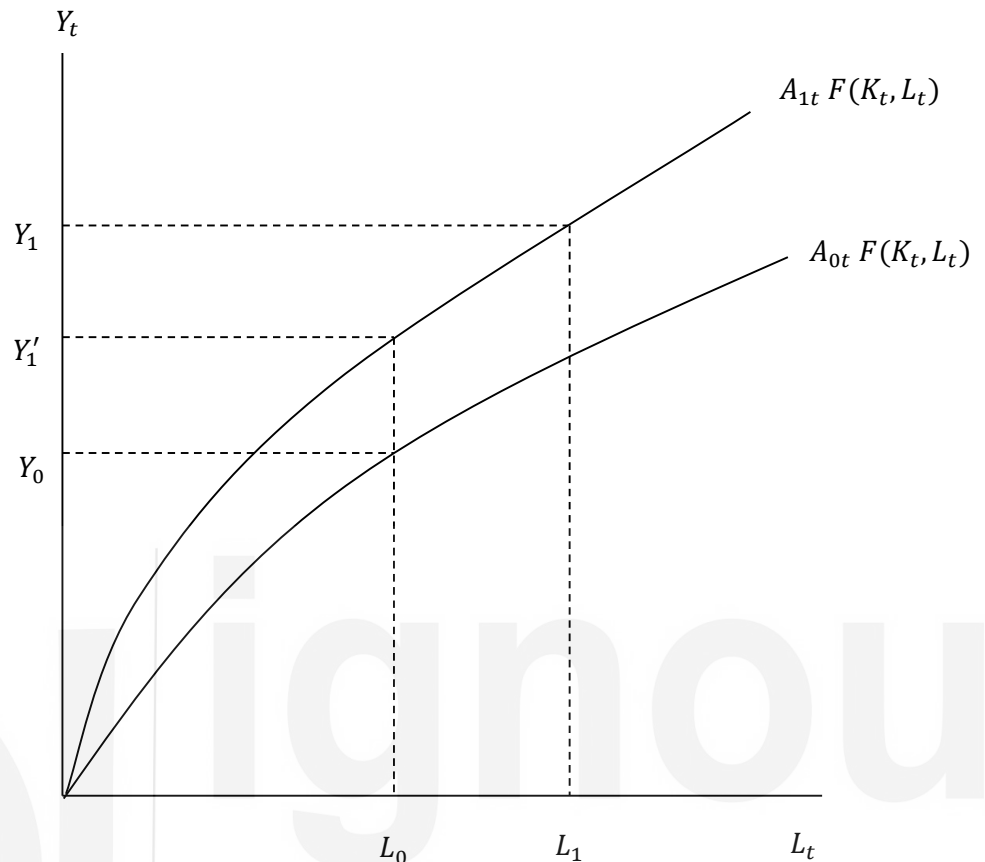


Fig. 12.2: Effect of Positive Technology Shock

Now, let us discuss about the distribution of such increased output. The increased output is distributed between consumption and saving. One possibility is that the increased output goes completely towards consumption. But, particularly in the case of a temporary shock, we may not prefer to consume all, but we like to save a portion of them for future higher consumption. If this is the case, equation (12.2) suggests that the higher the saving higher will be the level of investment. This will cause the capital stock to be higher in the next period as well as other future periods. Thus, the exogenous and temporary technology shock leads to increases in future capital stock.

Had the shock lasted for several periods or caused permanent effects, the economy would have responded somewhat differently. A permanent shock causes higher output for a longer period. Hence, the incentive to save would be reduced and consumption would be increased. The other possibility in the permanent shock case is that the increase in the labour input may not be high. Nevertheless, a persistent productivity shock will cause an increase in output, capital stock, and employment.

You should note that there is persistence of the effect of technology shocks. However, this statement is criticized by Keynesian economists as it cannot explain the persistence of real-world business cycles. The believers of RBC model argue that as explained in the model, the dynamic responses of optimizing

agents to change in economic conditions will have sustained effects. Such responses can explain periods of persistently high or low economic activity.

12.7 NEW-KEYNESIAN VIEW ON BUSINESS CYCLE

An important assumption of new-classical models is continuous market clearing, which implies perfect competition and flexibility in wages and prices. Thus, adjustments in prices and wages were instantaneous. Recall that classical economics was based on the same assumptions. Keynes discarded such assumptions and stressed on market imperfections. He advocated an activist role for the government.

A major criticism of Keynesian economics by Robert Lucas was that Keynesian economics does not have microfoundation. Further, expectations about future prices are not taken into account in Keynesian economics. New-Keynesian macroeconomics overcomes the above criticisms by including microfoundation and rational expectations.

What will happen if the above two features are brought into Keynesian economics? It implies that New-Keynesian economics also uses DSGE models. However, New-Keynesian models retain market imperfections and rigidities in prices and wages. Thus, there is not much difference between new-classical macroeconomics and new-Keynesian macroeconomics so far as methodology is concerned. The assumptions about market clearing are different however. New-Keynesian economists see certain imperfections in the market, particularly imperfect competition in the market and rigidities in prices and wages. In order to bring in such imperfections to DSGE models, new-Keynesian economists include various price-setting and wage-setting equations. These models take care of adjustment costs (such as menu costs) in wage setting, mark-up in price setting (as a measure of market imperfection) and some form of Taylor Rule for interest rate setting (policy rule).

Check Your Progress 3

- 1) Explain the impact of a supply shock on the economy through a diagram.

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- 2) State the issues emphasised by new-Keynesian economics in dynamic stochastic general equilibrium models.

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12.8 LET US SUM UP

Real business cycle (RBC) theory is based on microeconomic foundations in the sense that it takes into account the behaviour of economic agents such as firms and households. Thus, it considers the production function for analyzing changes in behavior of firms during a business cycle. Similarly, it considers the consumption function, particularly substitutability between labour and leisure on the part of households. Further, RBC theory is based on rational expectations in the sense that economic agents optimize on their objective functions by taking into account all information available to them. The RBC models adopt a general equilibrium framework for empirical estimation. The assumptions of RBC models, particularly its assumption of continuous market clearing, are challenged by new-Keynesian economics. The new-Keynesian economists, while adopting microfoundation, rational expectations and general equilibrium framework, incorporate market imperfections such as adjustment costs, staggered prices and mark-up pricing in dynamic stochastic general equilibrium (DGSE) models.

12.11 ANSWER/HINTS TO CHECK YOUR PROGRESS EXERCISES

Check Your Progress 1

- 1) There are three basic assumptions in new-classical models, viz., rational expectations, continuous market clearing, and Lucas supply function. You should discuss the implications of these assumptions. Refer to Section 12.2 for details.
- 2) There are basically reasons, viz., inter-temporal substitution between labour and leisure, and Lucas supply function. According to new-classical economics, business cycles are rational response of economic agents to shocks to the economy. Refer to Section 12.2.

Check Your Progress 2

- 1) There could be several examples of real shocks. The negative shocks for example could be decline in technology (i.e., technical regress), unexpected rise in input prices (crude oil crisis), scanty rainfall (severe drought), and large-scale increase in taxes on inputs. Refer to Section 12.3.
- 2) Refer to Section 12.4 and explain on the basis of equation (12.21).
- 3) Refer to equation (12.21). You can find that the ratio of labour supply in both the periods is influenced by the ratio of wage rates.
- 4) Refer to Section 12.4 and explain on the basis of equation (12.21).

Check Your Progress 3

- 1) Refer to Fig. 12.2 and answer.
- 2) Refer to Section 12.7 and answer.